

## Mariel Wynne de Vries Jones, PhD

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|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| ACADEMIC<br>BACKGROUND     | <i>Ph.D. Water Resources Science (Hydrology)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | May 2025       |
|                            | <a href="#">University of Minnesota</a> , Twin Cities, MN<br>Doctoral Dissertation Fellow and Provost's Award Recipient<br><i>Thesis:</i> Understanding Coupled Snow and Frost Behavior in Peatland Landscapes                                                                                                                                                                                                                                                                                                                                                                                       |                |
|                            | <i>B.Sc. General Engineering, Mathematics and Statistics</i><br><a href="#">Smith College</a> , Northampton, MA<br><i>Magna Cum Laude</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2020           |
|                            | <i>Junior Year Abroad</i><br><a href="#">University College Dublin</a> , Dublin, Ireland                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2019           |
| PROFESSIONAL<br>EXPERIENCE | <i>ORISE/USGS Climate Adaptation Consultant</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2025 - Present |
|                            | Midwest Climate Adaptation Science Center, Saint Paul, MN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                |
|                            | Leads research projects on hydrological impacts in the Midwest area. Main research project will evaluate weather hazards, including heavy rainfall and heatwaves, and assess the vulnerability of National Park Service (NPS) infrastructure including an assessment of future flood risk to culverts. In collaboration with NPS managers, provide information, analysis, and technical assistance to park managers to identify specific climate hazards, evaluate the risk to individual park assets, and make informed decisions in the maintenance or replacement of critical park infrastructure |                |
|                            | <i>Graduate Research Assistant</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2020 - 2025    |
|                            | Saint Anthony Falls Laboratory, Minneapolis, MN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                |
|                            | Research involves tracking changes in snow season lengths and modelling spring hydrological connectivity in peatland dominated areas of Minnesota with hopes to expand to the Midwest area. Work done in conjunction with the Saint Anthony Falls Laboratory (SAFL) in Minneapolis, MN and the USFS Marcell Experimental Forest Station in Grand Rapids, MN.                                                                                                                                                                                                                                         |                |
|                            | <i>Environmental Data Initiative (EDI) Fellow</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Summer 2021    |
|                            | Marcell Experimental Forest, Grand Rapids, MN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                |
|                            | Summer fellowship in collaboration with the Environmental Data Initiative (EDI) and the USFS Marcell Experimental Forest (MEF) to build data processing and publication knowledge within the ecological and environmental community. Participated in a weeklong training on data publication and worked for 8 weeks with the MEF site to process and publish long term meteorological and hydrological data.                                                                                                                                                                                         |                |
|                            | <i>Design Clinic Engineer</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2019 - 2020    |
|                            | Smith College Engineering Department, Northampton, MA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                |
|                            | Extensive collaboration with students and professionals on a project sponsored by the US Fish and Wildlife Service and the US Geological Survey. Work was done                                                                                                                                                                                                                                                                                                                                                                                                                                       |                |

to redesign entranceways to fish passages and increase both attraction and passage efficiencies from native fish species. 2 months were spent as Project Manager, and final deliverables were communicated to USFWS in the form of a design recommendation.

*Undergraduate Research Assistant* 2019 - 2020  
Smith College Engineering Department, Northampton, MA

Joint research between Smith College and the University of Massachusetts, Amherst in geotechnical engineering. Work involved testing railroad ballast and fouling material at different densities to characterize the stability of the material over time. Tests were done using both Lightweight Deflectometer and Ground Penetrating Radar techniques.

*Undergraduate Research Assistant* 2016 - 2018  
Smith College Mathematics Department, Northampton, MA

A year-round group research project partnering with Professor Julianna Tymoczko, and graduate-level researchers to explore the mathematical concept of Splines, graphical arrays of labeling often used in engineering design and modeling. Work includes both group collaboration and individual research to discover and prove properties of Splines.

## PUBLICATIONS See also [my google scholar](#) page.

### *Journal Publications (published)*

2. M.W. Jones, S. F. Dymond, S. D. Sebestyen, X. Feng, Forest Structural Diversity Increases Snow Accumulation in Low-relief Watersheds, *Hydrological Processes*, 2025. <https://doi.org/10.1002/hyp.70352>
1. M. W. Jones, S. D. Sebestyen, S. F. Dymond, G-H. C. Ng, X. Feng, Frost Decouples Spring Streamflow from Snowmelt in Headwater Catchments, *Journal of Hydrology*, 2023. <https://doi.org/10.1016/j.jhydrol.2022.128801>

### *Journal Publications (in prep)*

2. M. W. Jones, S. Swenson, S. D. Sebestyen, S. F. Dymond, X. Feng, Hillslope Representations of Peatland Catchments in the Community Land Model, *in prep for Journal of Advances in Modelling Earth Systems (JAMES)*
1. M.W. Jones, S. F. Dymond, S. D. Sebestyen, X. Feng, Trans-seasonal effects of soil frost on headwater catchments, *in prep for ARC Geophysical Research*

### *Data Publications*

3. M. W. Jones, G-H. C. Ng, S. D. Sebestyen, and X. Feng, *Marcell Experimental Forest 30-minute water table elevation and temperature from transects of wells in the S2 and S6 peatlands, 2018-ongoing*, Environmental Data Initiative (EDI), January 2024  
<https://doi.org/10.6073/pasta/4d1bce885d04c5c3ead0e682a016f968>
2. S.D. Sebestyen, D.T. Roman, J.M. Burdick, R.L. Kyllander, N.K. Lany, M. Jones, and R.K. Kolka, *Marcell Experimental Forest 15-minute precipitation, 2010 - ongoing*, Environmental Data Initiative (EDI), July 2021  
[doi:10.6073/pasta/73672ec2acdce8355bf8db](https://doi.org/10.6073/pasta/73672ec2acdce8355bf8db)
1. S.D. Sebestyen, D.T. Roman, N.K. Lany, M.W. Jones, J.M. Burdick, R.K. Kolka, *Marcell Experimental Forest 30-minute resolution meteorological data, 2006 - ongoing*, Environmental Data Initiative (EDI), August 2021  
<https://doi.org/10.6073/pasta/998c6c53ee>

*Refereed Conference Publications*

3. M. W. Jones, X. Feng, K. Hoffman, S. D. Sebestyen, S. F. Dymond Snow, soil frost, and hydrologic connectivity in peatland watersheds. *Proceedings from the 90th Annual Western Snow Conference*
2. E. K. Akey, M. W. Jones, C. L. Ho, A. J. Rubin, (2022) Measuring Railroad Ballast Modulus of Elasticity Using Light Weight Deflectometer. In: Tutumluer, E., Nazarian, S., Al-Qadi, I., Qamhia, I.I. (eds) *Advances in Transportation Geotechnics IV*. Lecture Notes in Civil Engineering, vol 165. Springer, Cham. <https://doi.org/10.1007/978-3-030-77234-522>
1. M. W. Jones, E. K. Akey, C. L. Ho, A. J. Rubin, (2022) Repeatability of Minimum and Maximum Density Testing on Clean and Fouled Ballast. In: Tutumluer, E., Nazarian, S., Al-Qadi, I., Qamhia, I.I. (eds) *Advances in Transportation Geotechnics IV*. Lecture Notes in Civil Engineering, vol 165. Springer, Cham. <https://doi.org/10.1007/978-3-030-77234-521>

*General Audience Publications*

2. M.W. Jones (2024). Please let it snow! Please let it snow! –Tracking changes in snow dynamics in northern Minnesota watersheds, University of Minnesota Water Resources Center Minnegramp Newsletter,
1. M. W. Jones (2021). Not so Renewable: Implications for continued peat mining in Minnesota. University of Minnesota Digital Conservancy, <https://hdl.handle.net/11299/229964>

**CONFERENCE** *Oral Presentations*

- PRESENTATIONS**
7. M. W. Jones, S. Swenson, S. D. Sebestyen, S. Dymond, X. Feng, Modeling Peatland Watersheds Using Hillslope Representations in CLM 5.0, *Computational Methods in Water Resources*, Tuscon, AZ, September 2024
  6. M. W. Jones, S. Dymond, S. D. Sebestyen, X. Feng, Forest Architecture Controls on Snow in Mississippi Headwater Catchments, *WaterSciCon*, Saint Paul, MN, June 2024
  5. M. W. Jones, X. Feng, K. Hoffman, S. D. Sebestyen, S. Dymond, Snow, soil frost, and hydrologic connectivity in peatland watersheds, *Western Snow Conference 2023*, Flagstaff, AZ, April 2023
  4. M. W. Jones, H. Anurag, G-H. C. Ng, X. Feng, A new frost-centric snowmelt partitioning framework — with applications to northern Minnesota forests, *AGU 2022 Fall Meeting*, Chicago, IL, December 2022
  3. M. W. Jones, S. D. Sebestyen, S. Dymond, X. Feng, The Role of Precipitation, Vegetation, and Evapotranspiration in Lowland, Snow-dominated Headwater Catchments, *AGU 2021 Fall Meeting*, New Orleans, LA (Virtual), December 2021
  2. E. K. Akey, M. W. Jones, C. L. Ho, A. J. Rubin, Measuring Railroad Ballast Modulus of Elasticity Using Light Weight Deflectometer, *International Conference on Transportation Geotechnics*, Chicago, IL (Virtual), August 2021
  1. M. W. Jones, E. K. Akey, C. L. Ho, A. J. Rubin, Repeatability of Minimum and Maximum Density Testing on Clean and Fouled Ballast, *International Conference on Transportation Geotechnics*, Chicago, IL (Virtual), August 2021

*Invited Talks*

1. Determining the Influence of Seasonally Frozen Ground in Minnesota Peatlands, *University of Minnesota Water Resources Science Seminar*, November 2022

#### Poster Presentations

6. M. W. Jones, S. F. Dymond, S. D. Sebestyen, X. Feng, Forest Structure Controls on Snow Accumulation in Headwater Catchments, *American Geophysical Union Fall Meeting*, Washington DC, DC, December 2024
5. S. Mohandes-Samani, M. W. Jones, F. Liu, S. D. Sebestyen, S. F. Dymond, X. Feng, Snowmelt Regulates Source Water Chemistry and Contributions to Streamflow in a Northern Forested Peatland Watershed, *American Geophysical Union Fall Meeting*, Washington DC, DC, December 2024
4. M. W. Jones, S. Swenson, S. D. Sebestyen, S. F. Dymond, X. Feng, Representing Peatland Heterogeneity in CLM5.0 Using Subgrid Hillslope Methods, *American Geophysical Union Fall Meeting*, San Francisco, CA, December 2023
3. M. W. Jones, S. F. Dymond, S. D. Sebestyen, X. Feng, Spatial Snow and Frost Patterns in Minnesota Peatland Watersheds, *Minnesota Water Resources Conference*, Saint Paul, MN, October 2023
2. M. W. Jones, X. Feng, Effects of Soil Frost on Streamflow Generation Processes in Minnesota Headwater Catchments, *Minnesota Water Resources Conference*, Saint Paul, MN, October 2022
1. X. Feng, M. W. Jones, G-H C. Ng, S. D. Sebestyen, Hydrology, Biogeochemistry & CH<sub>4</sub> Emissions at the Marcell Experimental Forest, *Department of Energy Environmental System Science PI Meeting*, Saint Paul, MN, July 2021 (Virtual)

#### AWARDS

- University of Minnesota, Doctoral Dissertation Fellowship, given to top PhD candidates across the university each year, 2024-2025
- Saint Anthony Falls Laboratory, Alvin Anderson Award for graduate work in water resources engineering, 2024
- AGU Catchment Hydrology Visual Abstract Competition, Winner, 2023
- University of Minnesota Provost's Award, given to top graduate school applicants, 2020
- Adeline Devor Penberthy Memorial Prize, excellence in engineering and fostering a community of learners, 2020
- Phi Beta Kappa (Academic Honor Society), Elected 2020
- Tau Beta Kappa (Engineering Honor Society), Elected 2019
- Smith College Dean's List, 2016-2018, 2020

#### GRANTS AND FELLOWSHIPS

- (2024-2025) University of Minnesota Doctoral Dissertation Fellowship for top PhD candidates, \$25,000 Annual Funding + \$1,000 Travel Grant
- (2022 - 2025) NSF Hydrological Sciences Grant #2153802 *Forest, Frost, and Flow: Snow Hydrology of Spatially Heterogeneous and Hydrologically Connected Peatland Catchments*, \$505,165, Supporting personnel under PIs Drs. Xue Feng (UMN-TC) and Salli Dymond (UMN-D, NAU).
- (2022 - 2024) Watershed Innovations Grant (WINS) *Snow Hydrology in Minnesota Headwater Catchments*, \$10,000, with Dr. Xue Feng (UMN-TC)
- (2020 - 2024) Water Resources Science Graduate Travel Funding, cumulatively awarded over three years, \$1,373
- (Fall 2023) Young Voices of Science (YVoS) Fellowship sponsored by Hubbard Brook Research Foundation

- (Fall 2023) Frank and Julie Tsai Travel Award for funding to attend the American Geophysical Union Fall Meeting, \$500
- (Summer 2021) Environmental Data Initiative Fellowship with Marcell Experimental Forest, \$5,000
- (Summer 2019) Undergraduate Research Fellowship, *Characterizations of Connecticut Granite Ballast materials and implications to USA High Speed Rail Networks*, \$5,000, under Dr. Aaron Rubin (Smith)
- (2016 - 2018) STRIDE Undergraduate Research Scholarship, \$45,000, Awarded to 50 (6-7%) incoming Smith College students annually

## CERTIFICATIONS

- FE General Exam, Passed January 2021
- Group Facilitation Skills Course, State of Minnesota Management and Budget Office, October 2025

## TECHNICAL SKILLS

*Software and Programming* C, Python, Java, R, MATLAB, Fortran, ArcGIS, Git

*Language Abilities* French (fluent), Italian (Beginner)

## LEADERSHIP

*Water Resources Science in Action Co-President* 2021 - 2022  
University of Minnesota, Minneapolis, MN

Student leadership position that helps to create and maintain a cohesive and friendly environment within an interdisciplinary program with students from colleges across both Twin Cities and Duluth campuses. Additionally, served as the student representative to the WRS Executive Committee.

*Picker Engineering Department Student-Faculty Liaison* 2019 - 2020  
Smith College, Northampton MA

Work done to help connect the department administration and student body by attending faculty meetings, hosting lunch talks for students and helping to guide new students through the engineering curriculum.

## TEACHING (at UMN)

- *CEGE 4501/5501 Hydrologic Design* Spring 2022  
Course development TA. CEGE 4501 is a core undergraduate class on the fundamentals of hydrologic analysis and an introduction to ecohydrology, stormwater engineering, and spatial analysis.
- *CEGE 3102 Uncertainty and Decision Analysis* Fall 2022  
Grading TA. This course is a sophomore level introduction to statistical analysis, probability theory, and their respective applications to uncertainty in design engineering.